SOFTWARE PROJECT MANAGEMENT

Week 5

Activity Planning & Scheduling

From WBS and effort estimates to a defensible schedule baseline

WBS	Activities	Dependencies	CPM / PERT	Gantt	Baseline
What?	Work?	Logic?	When?	Visualize	Control

Chapter focus: Activity Planning / Scheduling

COURSE MAP

Where Week 5 Fits



Week 5 question

Given the approved scope and effort estimate, what sequence of activities determines the earliest defensible completion date?

OBJECTIVES

Learning Outcomes

- Transform WBS work packages into schedulable activities and milestones.
- Model dependency logic using FS, SS, FF and SF relationships.
- Use leads and lags only when the timing logic is defensible.
- Build Activity-on-Node networks and perform forward/backward passes.
- Calculate float and identify the critical path.
- Use PERT when duration estimates are uncertain.
- Translate the network into a Gantt chart and schedule baseline.

Today’s teaching pattern

Worked example → short problem → solution → next concept.

BIG PICTURE

The Scheduling Chain



Key message

Project duration is created by dependency logic — not by simply adding all activity durations.

TRY IT

Problem 1 — Can We Finish in 8 Weeks?

Mini case

Requirements = 2 weeks • UI Prototype = 2 weeks • Backend = 4 weeks • Testing = 2 weeks

Management adds the durations: 2 + 2 + 4 + 2 = 10 weeks.
Is 10 weeks necessarily the project duration?

Vote

A) Yes — duration is always the sum • B) No — some activities may overlap • C) Not enough information

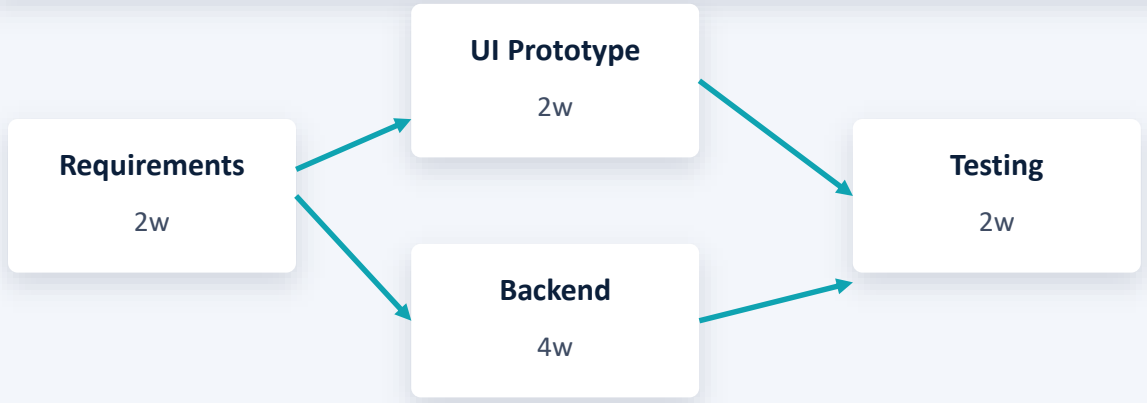
Think 30 seconds before answering.

REASONING

Problem 1 — Solution

Correct answer: C

We cannot determine the project duration until we know the dependency relationships.



Because UI and Backend can run in parallel after Requirements, the network — not the sum — determines the finish.

VOCABULARY

Six Scheduling Terms Students Must Separate

Activity

Consumes time and usually resources.

Milestone

Zero-duration significant event.

Effort

Total human work.

Duration

Calendar time.

Dependency

Logical relationship between activities.

Float

Allowable delay without affecting a specified date.

INPUTS

Scheduling Inputs: What Must Exist First?

- Approved scope and WBS from Week 3.
- Work-package definitions and acceptance boundaries.
- Effort and duration assumptions from Week 4.
- Known external dependencies: vendors, APIs, approvals, client reviews.
- Project calendars: working days, holidays and fixed dates.
- Contractual or business deadlines.
- Known technical sequencing constraints.

Do not schedule vague scope

If “Payment” could mean one API call or a multi-provider secure integration, the schedule cannot be credible.

Do not confuse effort with duration

160 person-hours may be 4 weeks for one person, 2.5 weeks for two people, or longer if work cannot be parallelized.

ABC RETAIL

Example — From WBS Work Package to Activities

WBS work package

1.6 Payment Integration

WBS answers WHAT

Activities answer HOW the deliverable will be produced.

1. Confirm gateway requirements
2. Obtain sandbox credentials
3. Configure SDK / API
4. Implement payment request
5. Handle callbacks & errors
6. Security review
7. Integration testing
8. Acceptance

QUALITY CHECK

What Makes a Good Activity?

Action-oriented

Begin with a verb: Configure, Develop, Review, Test, Approve.

Observable completion

Someone can tell when it is finished.

Appropriate size

Not so large that progress is invisible; not so small that the schedule is cluttered.

One clear owner

Responsibility can be assigned.

Dependency-ready

Predecessor/successor logic can be stated.

Estimate-ready

Duration can be estimated with a clear basis.

TRY IT

Problem 2 — Turn a Work Package into Activities

Smart Clinic WBS work package

1.5 Appointment Booking & Rescheduling

Write 5–7 activities that would be suitable for scheduling.

Hint 1

Use verbs

Hint 2

Avoid “do module”

Hint 3

Include review/test

Hint 4

Make completion observable

EXAMPLE SOLUTION

Problem 2 — One Good Answer

ID	Activity
A1	Confirm booking rules
A2	Design booking interface
A3	Implement slot search
A4	Implement booking/rescheduling
A5	Add validation and conflict checks
A6	Execute module tests
A7	Obtain product-owner acceptance

There can be other correct answers

What matters is that the activities are schedulable, observable and cover the work package.

MILESTONES

Milestones: Zero Duration, High Management Value

Requirements approved

0 days

Prototype accepted

0 days

Payment sandbox ready

0 days

System test complete

0 days

UAT sign-off

0 days

Go-live

0 days

Use milestones for

Approvals • decision points • contractual events • release checkpoints • handovers

QUICK CLASSIFICATION

Problem 3 — Activity or Milestone?

Item	Activity or Milestone?
Conduct requirements workshop	?
Requirements approved	?
Develop payment interface	?
UAT sign-off	?
Deploy release	?
Go-live decision approved	?

Rule: if it consumes work/time, it is usually an activity. If it represents an event/checkpoint with zero duration, it is a milestone.

DEPENDENCIES

Four Dependency Relationships

FS

Finish → Start
Successor starts after predecessor finishes.

SS

Start → Start
Successor starts after predecessor starts.

FF

Finish → Finish
Successor finishes after predecessor finishes.

SF

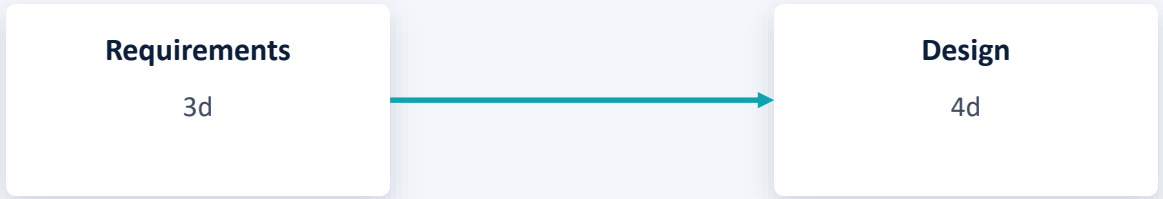
Start → Finish
Successor finishes after predecessor starts.

Most software schedules

Use FS most often. SS and FF are useful for controlled overlap. SF is rare.

EXAMPLE

Finish-to-Start (FS) — The Default



Design cannot start until Requirements finishes.

Common software examples

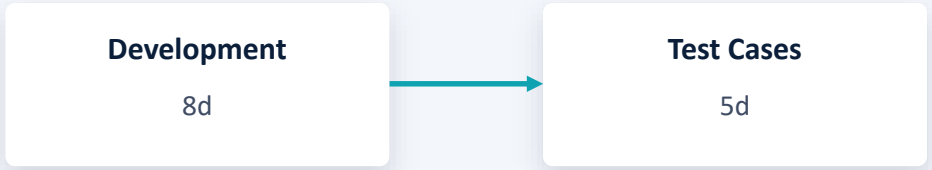
Requirements → Design • Design → Build • Build → System Test • UAT → Go-Live

EXAMPLES

Start-to-Start and Finish-to-Finish

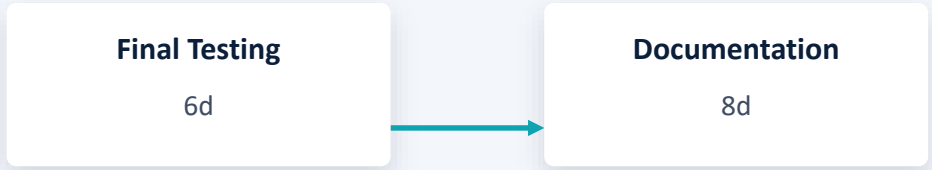
SS example

Test-case preparation may start once development starts. The two activities overlap.



FF example

User documentation must finish no earlier than final testing finishes.



SS controls start timing. FF controls finish timing.

EXAMPLE

Start-to-Finish (SF) — Rare but Real

Support handover

The old support shift cannot finish until the new support shift starts.



Use SF only when the real-world logic genuinely requires it.

TRY IT

Problem 4 — Choose the Relationship

Scenario	FS / SS / FF / SF?
Design must be approved before coding starts.	?
Test-case writing may begin when development starts.	?
User documentation must not finish before final testing finishes.	?
Old support shift cannot finish until new shift starts.	?

Discuss with a partner before revealing the solution.

ANSWER

Problem 4 — Solution

Scenario	Relationship	Reason
Design approved before coding starts	FS	Coding waits for design completion.
Test-case writing begins when development starts	SS	Both can overlap after development starts.
Documentation cannot finish before final testing	FF	Finish timing is linked.
Old support ends after new support starts	SF	New shift start controls old shift finish.

TIMING OFFSETS

Leads and Lags

Lag = intentional waiting

Example: UAT starts 2 days after System Testing finishes → FS + 2d

Lead = controlled overlap

Example: Training material begins 2 days before final testing finishes → FS − 2d

Do not use leads/lags to hide weak logic.

Ask first

Can the activities truly overlap? Is the waiting time mandatory? What rework risk does the overlap create?

TRY IT

Problem 5 — Add the Correct Offset

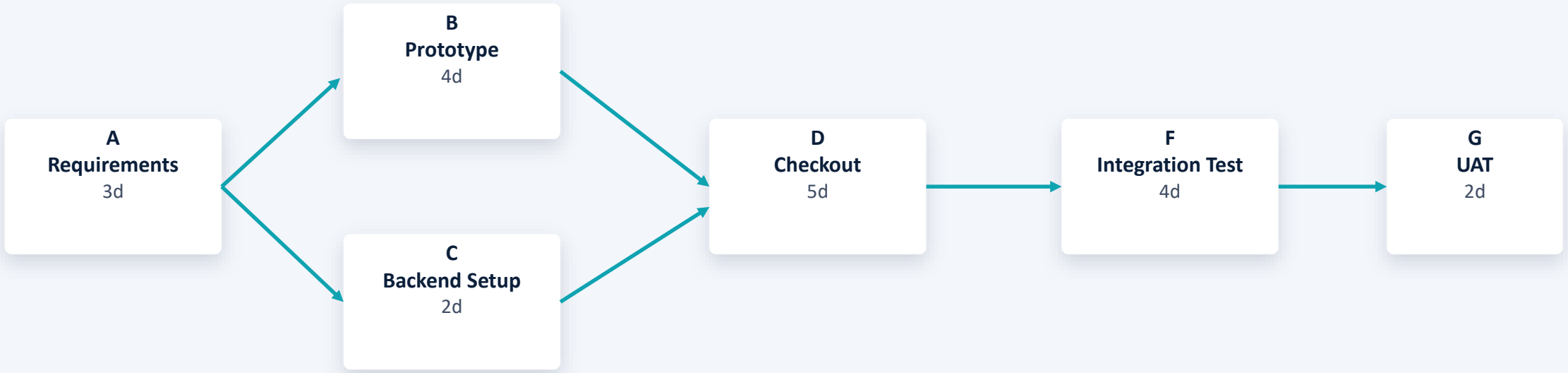
Scenario	Relationship notation
UAT begins 3 days after System Testing finishes.	?
Training material starts 2 days before final testing finishes.	?
Security review starts at the same time as integration testing.	?

Possible answers

FS + lag • FS − lead • SS

NETWORK DIAGRAM

Activity-on-Node (AON) Network



Read left to right

Nodes = activities. Arrows = dependency logic. The network makes parallelism and merge points visible.

SMART CLINIC

Problem 6 — Build the Network

ID	Activity	Duration	Predecessor(s)
A	Requirements	2d	—
B	Prototype	3d	A
C	Doctor Schedule	4d	A
D	Appointment Module	5d	B, C
E	Queue Module	3d	C
F	Integration Test	4d	D, E
G	UAT	2d	F

Draw the AON network before calculating dates.

CPM

Forward Pass — Earliest Start and Finish

Formula 1

EF = ES + Duration

Formula 2

ES(successor) = MAX[EF of all predecessors]

Why MAX?

A successor with multiple predecessors must wait until all required predecessor work is ready. Therefore, it starts after the latest predecessor finish.

Use ES = 0 for the first activity in our classroom calculations.

WORKED EXAMPLE

Forward Pass — Worked ABC Retail Example

ID	Dur	Pred	ES	EF
A	3	—	0	3
B	4	A	3	7
C	2	A	3	5
D	5	B,C	7	12
E	3	C	5	8
F	4	D,E	12	16
G	2	F	16	18

Project earliest finish

18 days

SMART CLINIC

Problem 7 — Calculate the Forward Pass

ID	Dur	Pred	ES	EF
A	2	—		
B	3	A		
C	4	A		
D	5	B,C		
E	3	C		
F	4	D,E		
G	2	F		

What is the earliest project finish?

ANSWER

Problem 7 — Solution

ID	Dur	Pred	ES	EF
A	2	—	0	2
B	3	A	2	5
C	4	A	2	6
D	5	B,C	6	11
E	3	C	6	9
F	4	D,E	11	15
G	2	F	15	17

Earliest finish

17 days

CPM

Backward Pass — Latest Start and Finish

Formula 1

LS = LF − Duration

Formula 2

LF(predecessor) = MIN[LS of all successors]

Why MIN?

A predecessor with several successors must finish early enough to avoid delaying the earliest required successor.

Start from the project earliest finish and move right → left.

WORKED EXAMPLE

Backward Pass — Worked ABC Retail Example

ID	Dur	LS	LF
A	3	0	3
B	4	3	7
C	2	5	7
D	5	7	12
E	3	9	12
F	4	12	16
G	2	16	18

Notice

C and E have scheduling flexibility. Their late dates are later than their early dates.

SMART CLINIC

Problem 8 — Calculate the Backward Pass

ID	Dur	ES	EF	LS	LF
A	2	0	2		
B	3	2	5		
C	4	2	6		
D	5	6	11		
E	3	6	9		
F	4	11	15		
G	2	15	17		

Begin with G: LF = 17.

ANSWER

Problem 8 — Solution

ID	Dur	ES	EF	LS	LF
A	2	0	2	0	2
B	3	2	5	3	6
C	4	2	6	2	6
D	5	6	11	6	11
E	3	6	9	8	11
F	4	11	15	11	15
G	2	15	17	15	17

FLOAT

Total Float — How Much Delay Can We Absorb?

Formula

Total Float = LS − ES = LF − EF

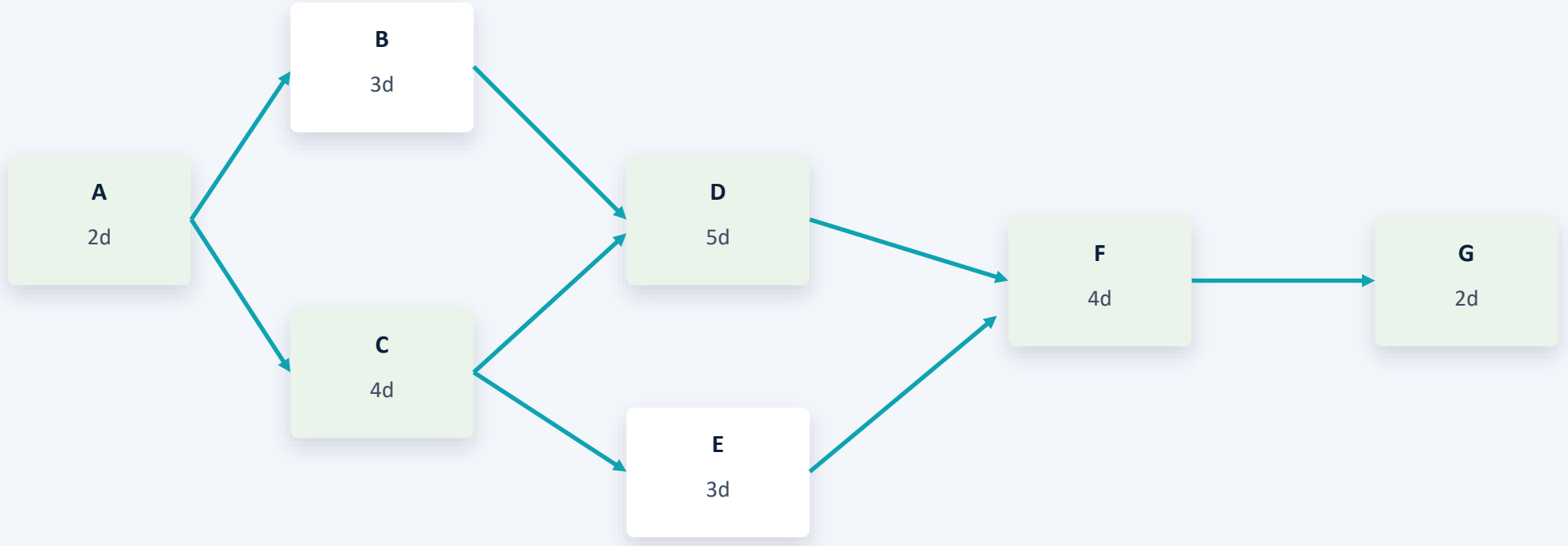
Activity	ES	LS	Float
A	0	0	0
B	2	3	1
C	2	2	0
D	6	6	0
E	6	8	2
F	11	11	0
G	15	15	0

Interpretation

B can slip 1 day and E can slip 2 days without delaying the current project finish — assuming all other logic stays unchanged.

CPM

Critical Path — The Path That Controls the Finish



Critical path

A → C → D → F → G = 17 days

WORKED EXAMPLE

ABC Retail CPM — Complete Calculation

ID	Dur	Pred	ES	EF	LS	LF	Float
A	3	—	0	3	0	3	0
B	4	A	3	7	3	7	0
C	2	A	3	5	5	7	2
D	5	B,C	7	12	7	12	0
E	3	C	5	8	9	12	4
F	4	D,E	12	16	12	16	0
G	2	F	16	18	16	18	0

Critical path

A → B → D → F → G = 18 days

TRY IT

Problem 9 — Identify the Critical Path

Activity	Duration	Predecessor(s)
A	2	—
B	5	A
C	4	A
D	3	B
E	6	C
F	2	D,E

Questions: 1) What is the project duration? 2) What is the critical path? 3) Which branch has float?

ANSWER

Problem 9 — Solution

Path 1: A → B → D → F = 2 + 5 + 3 + 2 = 12 days

Path 2: A → C → E → F = 2 + 4 + 6 + 2 = 14 days

Critical path

A → C → E → F

Project duration

14 days

Noncritical branch

B–D path has 2 days of float.

WATCH OUT

Critical Path — Common Misconceptions

Statement	Correct?	Why?
“Critical means most important.”	No	It means zero float under the current network.
“There is always one critical path.”	No	Multiple paths can tie for the longest duration.
“Critical activities can never change.”	No	The path can change as actual progress and logic change.
“Add people and the critical path disappears.”	No	Dependencies and work structure still matter.

THREE-POINT ESTIMATION

PERT — When Duration Is Uncertain

Optimistic (O)

Best credible duration if things go unusually well.

Most Likely (M)

Expected duration under normal conditions.

Pessimistic (P)

Credible duration if major difficulties occur.

Expected duration

$$TE = (O + 4M + P) / 6$$

PERT gives more weight to the most-likely estimate while still reflecting uncertainty.

WORKED EXAMPLE

PERT Worked Example — Payment Integration

Estimate	Value
Optimistic (O)	4 days
Most Likely (M)	7 days
Pessimistic (P)	13 days

Expected duration

$(4 + 4\times7 + 13) / 6 = 45 / 6 = 7.5$ days

Standard deviation

$(P - O) / 6 = (13 - 4) / 6 = 1.5$ days

Variance

$1.5^2 = 2.25$ days^2

TRY IT

Problem 10 — PERT Calculation

Smart Clinic notification service

O = 2 days • M = 5 days • P = 8 days

Calculate:

- Expected duration (TE)
- Standard deviation
- Variance

Formula reminder

TE = (O + 4M + P) / 6
SD = (P − O) / 6
Variance = SD^2

ANSWER

Problem 10 — Solution

Expected duration

$(2 + 4\times5 + 8) / 6 = 30 / 6 = 5$ days

Standard deviation

$(8 - 2) / 6 = 1$ day

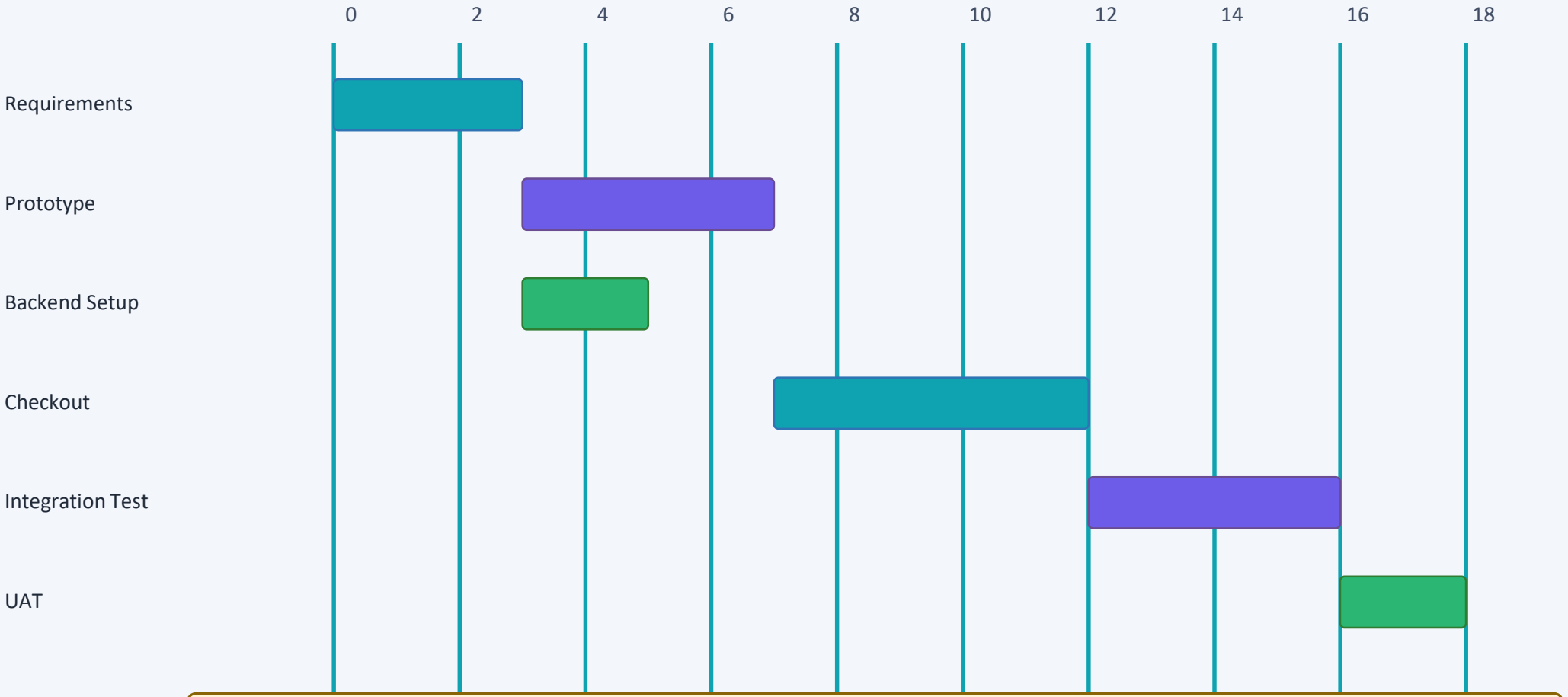
Variance

$1^2 = 1$ day^2

The expected duration equals the most-likely value here because the optimistic and pessimistic values are symmetric around M.

VISUALIZATION

Gantt Chart — Make the Network Visible



Gantt is not a substitute for logic

The bar chart communicates the schedule; the network/dependencies create it.

CONTROL FOUNDATION

Schedule Baseline vs Current Schedule

Schedule Baseline

The formally approved schedule used as the reference for later performance measurement. It should not move every time actual work changes.

Current / Forecast Schedule

The latest view of what is actually happening and when the project is now expected to finish.

Why it matters

Without a stable baseline, Week 11 monitoring cannot calculate or explain schedule variance meaningfully.

MANAGEMENT JUDGMENT

Problem 11 — Should We Change the Baseline?

Scenario

The Checkout activity is 3 days late. The project manager edits the baseline so the schedule again appears on time.

Is this acceptable?

Answer

No. Update the current/forecast schedule, investigate the variance and take corrective action. Change the baseline only through an approved control process when the authorized plan genuinely changes.

TOOL TRANSFER

From Manual Logic to Microsoft Project

1. Enter activities

Use the WBS hierarchy and activity list.

2. Enter durations

Use agreed duration assumptions.

3. Link predecessors

FS / SS / FF / SF + justified offsets.

4. Review critical path

Compare with your manual result.

5. Add milestones

Approvals and checkpoints.

6. Save baseline

Only after review/approval.

PRACTICAL

Week 5 Practical Deliverables

- Activity List derived from the approved Week 3 WBS.
- Duration assumptions linked to Week 4 estimation.
- Predecessor/dependency table with justified relationships and offsets.
- Manual Activity-on-Node network diagram.
- Forward pass and backward pass calculations.
- Total float and identified critical path.
- PERT calculation for at least one uncertain activity.
- Milestone list, Gantt chart and Microsoft Project schedule baseline.

Lab rule

Manual CPM first → Microsoft Project second → compare the results.

Do not do yet

Resource leveling, crashing and detailed resource optimization belong mainly to Week 6.

QUICK QUIZ

Final Challenge — Can You Defend the Schedule?

- 1. Why can project duration be shorter than the sum of all activity durations?
- 2. Which relationship means successor starts after predecessor finishes?
- 3. Why do we use MAX predecessor EF in the forward pass?
- 4. What does total float = 0 mean?
- 5. Can there be two critical paths?
- 6. What is the PERT expected-duration formula?
- 7. Why should the schedule baseline remain stable?

Week 5 takeaway

A professional schedule is a logical model of the work — not just a list of dates.